

of cryptographic literature. For the record, the alphabet rotation technique in the beginning is an example of encoding/ decoding.

To avoid confusion, for now let's assume that the terms are interchangeable.

(Cryptographic) Algorithm

In computer science, algorithms are a set (series) of instructions which when executed would solve a problem. The word retains its meaning in cryptography. The algorithm one uses to encrypt / decrypt messages is called 'cryptographic algorithm'. Note that when we talk about cryptographic algorithms, there will always be a pair. The first algorithm of the pair is responsible for encryption of the data i.e. to convert plaintext to ciphertext and the second one is responsible for decryption i.e. to convert the ciphertext back to plaintext. Though not in all, but in many cases (typically symmetric algorithms) the decryption algorithm is basically the reverse of the encryption algorithm and vice versa. For this reason, in most literary works on the science of cryptography, both the algorithms are referred to as one algorithm only. They're usually also implemented as a single program, hence the style of reference.

Encryption Key

Encryption or ciphering requires two things – an algorithm and at least one key. The algorithm converts the plaintext into ciphertext based on both, the algorithm as well as the key. If you change any one of them, the result will differ. Taking again the alphabet rotation example, the algorithm is 'Forward Rotation' and the 'Key' is '1'. If we would have rotated the alphabet by 2 places (e.g. replace A with C, B with D and so on), then the key would have been '2'.

Note that if you change the algorithm (like from forward rotation to backward rotation) or the key (e.g. rotate 3 places or 7 places) then the ciphertext will change. A good algorithm and a strong key are the necessary properties to make an encryption system strong.

Symmetric Ciphers

'Symmetric ciphers' or 'symmetric algorithms' are algorithms which use the same key to encrypt as well as decrypt data. Traditionally, this type of algorithm has been in use in methods of cryptography. The example that we gave is one of a symmetric cipher. When you encrypt the plaintext by replacing each letter with a letter one place ahead in the alphabet then you